Reframing Text Simplification Evaluation: a Retrieval-Based Perspective*

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Abstract

Text simplification eases document understanding while preserving meaning, but this balance remains difficult to evaluate without reference simplifications. We propose a retrieval-based perspective: if a simplified document preserves the information content of its source, it should remain retrievable for the same queries. We test this idea on BEIR collections from the biomedical, news, and Wikipedia domains, comparing original and simplified documents under BM25 and dense retrieval, and relating retrieval effectiveness and ranking stability to indicators of faithfulness and simplification quality. Simplification substantially affects retrieval behavior, especially when models aggressively compress or delete content, while conservative simplifiers better preserve search effectiveness. Search behavior thus provides a practical, reference-free signal for evaluating meaning preservation in text simplification.

Keywords

Text Simplification, Meaning Preservation, Retrieval Effectiveness, Reference-free Evaluation

1. Introduction

Text simplification (TS) aims to rewrite a source text into a more accessible form while preserving its original meaning, thereby supporting readers with different levels of literacy, domain expertise, or cognitive needs. Text simplification has evolved from rule-based methods [2, 3, 4] to supervised approaches trained on parallel data, including monolingual machine translation [5] and neural sequence-to-sequence models [6, 7], and, more recently, to LLM-based prompting [8, 9, 10, 11]. Despite this evolution in how simplification systems are developed, their evaluation remains challenging.

Assessing TS requires jointly evaluating simplicity and meaning preservation, but existing metrics have different limitations. Reference-based metrics such as SARI [12] and LENS [13] require gold simplifications, which are costly and unavailable for most domains, while reference-free measures only partially capture these two dimensions: readability indices [14] describe surface properties, and semantic similarity measures [15] overlook omissions and other forms of information loss [16]. These limitations have motivated more targeted evaluation methods, including QA-based approaches that assess whether source information remains recoverable [17, 18]. Document-level evaluations further show that simplification systems rarely perform well on both dimensions, and that existing metrics fail to track this tension [19].

We address this gap from an information retrieval perspective. Our hypothesis is that a simplified document preserving the relevant information of its source should remain retrievable for the same information needs. Retrieval can thus provide an indirect, reference-free signal of meaning preservation, grounded in a downstream task. This relates to QA-based evaluation, but, rather than testing whether

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individual facts remain recoverable, we ask whether the document as a whole still satisfies the same queries.

We investigate three research questions: (RQ1) Does text simplification affect retrieval behavior? (RQ2) Do simplification models differ in terms of simplicity and meaning preservation? (RQ3) Can retrieval behavior serve as a proxy for meaning preservation in text simplification?

Results show that simplification affects retrieval unevenly, depending on both the simplification model and the target domain, and that meaning-preservation metrics correlate significantly with retrieval effectiveness in stratified analyses. Our contributions are twofold: we propose the first retrieval-inspired evaluation perspective for text simplification, and we introduce retrieval analysis as a complementary tool for assessing meaning preservation.

2. Retrieval-Based Evaluation Framework

We frame text simplification as a retrieval-preservation problem. Given an original collection $\mathscr{D}$ and a simplified collection $\mathscr{D}'$, obtained by simplifying each document independently, we ask whether queries that retrieve a document from $\mathscr{D}$ still retrieve its simplified counterpart with comparable effectiveness and ranking behavior. No reference simplifications are needed, since the original collection itself provides the term of comparison. Retrieval works as an indirect signal because deletions, distortions or hallucinations introduced during simplification change whether a document still matches the same information needs. We therefore compare the two collections under identical query sets and retrieval models, measuring effectiveness and ranking stability in both lexical and semantic scenarios.

3. Experimental Setup

Simplification models. We use publicly available document-level simplification models [20, 21], as reproduced and extended by [22]: LED_{para}, a paragraph-level Longformer encoder-decoder model, and $\hat{O} \rightarrow$ ConBART, a plan-guided sentence-level pipeline with document-level context attention. For each we evaluate the variants trained on Newsela [21], Wiki-auto [22] and Cochrane-auto [23], covering the journalistic, encyclopedic and biomedical domains, running inference with the original authors' parameters. As a content-agnostic lower bound we include an FKGL-threshold baseline ($Baseline_{FKGL}$) that extracts at most three sentences with the lowest FKGL scores in their original order.

Datasets and retrieval pipeline. We use *NFCorpus* (biomedical), *Robust04* (news) and *NQ* (Wikipedia) from BEIR [24], chosen to match the domains above and thus to cover both in-domain and out-of-domain conditions. We build separate indices for the original collection and for each simplified version, querying all of them with the unchanged BEIR queries. We use BM25 as a sparse model and, in the dense setting, encode queries and documents with Snowflake[1], retrieving by inner product on a flat index built with `kANNolo` [25]; exact search avoids approximation error varying across corpora and confounding the comparison.

Metrics. We assess simplicity with FKGL, and meaning preservation against the source document with Length Ratio (compression), ROUGE-L F-measure [26] (lexical overlap), MeaningBERT (semantic similarity) [27], BERTScore [15] and BARTScore [28]. Retrieval and simplification are performed over the full collection, while simplicity and meaning preservation are evaluated on the documents appearing in the relevance judgments, providing a consistent evaluation set across simplified versions.

Due to space constraints, we report a subset of these measures, chosen to cover complementary aspects of the phenomenon: MAP and Recall@100, the retrieval metrics most affected by simplification in our experiments, together with Rank-Biased Overlap (RBO) [29], which captures ranking stability; on

[1] https://huggingface.co/Snowflake/snowflake-arctic-embed-l

the simplification side, we report ROUGE-L and MeaningBERT, which measure meaning preservation at the lexical and semantic level. The remaining metrics follow the same trends and are reported in [1].

4. Results and Discussion

Table 1 reports MAP, Recall@100 and RBO [29]; Table 2 reports ROUGE-L and MeaningBERT on relevant documents. We discuss them together, since preservation scores help explain the retrieval patterns we observe.

Table 1
Retrieval effectiveness (MAP, Recall@100) and top-10 ranking stability (RBO). OCon is $\hat{O}$ →ConBART. * marks a significant difference w.r.t. Original and † w.r.t. *Baseline*$_{FKGL}$ (paired t-test, Bonferroni correction, $p < 0.05$).

		NFCorpus			Robust04			NQ		
Model	Training data	MAP	R@100	RBO	MAP	R@100	RBO	MAP	R@100	RBO
BM25										
Original		**0.151**	**0.249**	–	**0.239**	**0.391**	–	0.239	**0.738**	–
Baseline$_{FKGL}$		0.109	0.191	0.17	0.063	0.146	0.06	0.155	0.598	0.37
LED$_{\text{para}}$	Newsela	0.125*†	0.206*†	0.36	0.104*†	0.212*†	0.16	0.198*†	0.665*†	0.31
	Cochrane-auto	0.144*†	0.243†	0.80	0.183*†	0.325*†	0.63	0.234*†	0.732*†	0.89
	Wiki-auto	0.150*†	0.247†	1.00	0.199*†	0.333*†	0.81	**0.241**†	0.737†	0.96
OCon	Newsela	0.147*†	0.243*†	0.85	0.166*†	0.301*†	0.37	0.171*†	0.631*†	0.19
	Cochrane-auto	0.148†	0.244†	0.81	0.224*†	0.371*†	0.76	0.210*†	0.681*†	0.55
	Wiki-auto	0.121*†	0.209*†	0.25	0.190*†	0.336*†	0.64	0.198*†	0.658*†	0.47
Flat Index (Snowflake)										
Original		**0.153**	**0.306**	–	**0.190**	**0.344**	–	**0.315**	**0.782**	–
Baseline$_{FKGL}$		0.056	0.147	0.01	0.031	0.094	0.06	0.183	0.487	0.08
LED$_{\text{para}}$	Newsela	0.079*†	0.197*†	0.01	0.065*	0.156*†	0.13	0.250*†	0.662*†	0.06
	Cochrane-auto	0.129*†	0.275*†	0.47	0.134*†	0.267*†	0.25	0.309*†	0.771*†	0.56
	Wiki-auto	0.150†	0.301*†	0.49	0.172*†	0.321*†	0.50	0.303*†	0.765*†	0.36
OCon	Newsela	0.078*†	0.212*†	0.24	0.119*†	0.259*†	0.22	0.187*	0.546*†	0.05
	Cochrane-auto	0.148*†	0.301†	0.49	0.160*†	0.322*†	0.38	0.278*†	0.699*†	0.15
	Wiki-auto	0.088*†	0.201*†	0.01	0.133*†	0.281*†	0.32	0.248*†	0.629*†	0.08

Retrieval Impact and Simplification Quality (RQ1 & RQ2). Simplification generally reduces both retrieval effectiveness and ranking stability, with larger effects for dense retrieval. This difference likely reflects how the two retrieval paradigms match documents. BM25 relies on retained terms, so preserving key terminology can maintain matching, whereas dense retrieval represents the document as a whole and is more sensitive to deletions and reformulations. NQ is the least affected corpus, likely because its shorter documents leave less content to change, while NFCorpus and Robust04 degrade more clearly. Most neural runs significantly outperform *Baseline*$_{FKGL}$, showing that the degradation cannot be attributed to content-agnostic sentence selection.

Table 2 shows a clear readability–preservation trade-off. Newsela produces the strongest simplification, but also the lowest lexical and semantic meaning preservation and larger retrieval degradation. Cochrane-auto is more conservative and preserves retrieval best across datasets. Wiki-auto under LED$_{\text{para}}$ remains closest to the source, with RBO@10 reaching 1.00 on NFCorpus, but its outputs are often nearly unchanged, limiting the actual simplification gain. The planner provides no consistent benefit: it improves Newsela and Cochrane-auto on Robust04, but can hurt on shorter documents such as NQ. Overall, retrieval reflects whether a model favors stronger simplification or greater preservation.

Retrieval and Meaning Preservation (RQ3). Aggregate Spearman correlations are weak, but stronger associations emerge at the extremes: we stratify queries into *high* and *low quality* groups

Table 2. Meaning preservation on relevant documents: ROUGE-L (R-L) and MeaningBERT (MB).

	NFCorp.		Rob04		NQ	
System	R-L	MB	R-L	MB	R-L	MB
$Baseline_{FKGL}$	0.24	53.9	0.15	50.2	0.48	70.9
LED-News	0.19	50.3	0.22	60.1	0.54	75.2
LED-Coch	0.82	87.1	0.68	86.9	0.99	93.9
LED-Wiki	0.97	93.1	0.76	90.9	0.97	93.6
OCon-News	0.29	59.9	0.49	78.1	0.51	71.6
OCon-Coch	0.87	90.0	0.85	90.3	0.76	85.6
OCon-Wiki	0.33	63.5	0.62	82.4	0.51	74.1

Table 3. Within-group Spearman correlations ($|\rho| \geq 0.40$, $p < 0.05$) with query-level MAP.

Run	R-L	MB
NFCorpus — high eff.		
dense-LED-News		
bm25-LED-News		
dense-LED-Coch	0.51	0.53
dense-LED-Wiki	**0.65**	**0.73**
Robust04 — low qual.		
dense-LED-News	0.47	
dense-LED-Wiki	0.44	

and, symmetrically, into *high* and *low effectiveness* groups, using the 25th and 75th percentiles as thresholds. On NFCorpus, the strongest correlations occur in the *high effectiveness* group, with ROUGE-L and MeaningBERT reaching 0.65 and 0.73 for Wiki-auto, whose near-identical outputs suggest these values mainly quantify how far highly preserved documents sustain retrieval. On Robust04, stronger associations appear for *low quality* simplifications, with lexical metrics showing the clearest effects, consistent with longer documents where retrieval depends more on surface-level overlap. Dense retrieval shows stronger correlations overall, consistent with its greater sensitivity to semantic changes. Retrieval therefore appears most informative when simplification is either highly disruptive or highly conservative.

5. Conclusions and Future Work

We cast text simplification as an information retrieval problem, asking whether simplified documents remain accessible under the same information needs as their originals. Across datasets and retrieval models, simplified collections reduce effectiveness and ranking stability, particularly under aggressive rewriting, indicating that current systems discard content relevant to information access rather than merely compressing it.

Two limitations bound these conclusions. Individual retrieval models may miss specific semantic phenomena — BM25 could be insensitive to negation, and synonym substitution reduces exact matching even when meaning is retained — so results should be read jointly across both sparse and dense paradigms. More fundamentally, retrieval captures topical relevance rather than factual accuracy, and human judgments are needed to establish whether it reflects aspects of preservation that existing measures miss; our results offer some indication that it does, since retrieval effectiveness can decrease while MeaningBERT remains high.

Future work will characterize the readability–retrieval trade-off directly, varying the degree of simplification to identify how much can be introduced before retrieval degrades substantially, so that retrieval can guide the design of simplification systems. We also plan to extend the analysis to learned sparse and late-interaction retrievers, and to LLM-based simplifiers.

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Declaration on Generative AI

Generative AI tools were used only to assist with language polishing and grammar checking. All research ideas, experimental design, implementation, analysis, and scientific claims were produced and critically reviewed by the authors.

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